



HAMMERTIME

V2

PRODUCT REQUIREMENTS DOCUMENT — FROM PREDICTION ENGINE
TO RACE STRATEGY COMMAND CENTER

DOCUMENT REF HT2-PRD-002	REVISION 2.0	ISSUED 14 JUN 2026
STATUS FOR REVIEW	CLASSIFICATION CONSTRUCTORS' EYES ONLY	PROGRAMME PITWALL → HAMMERTIME

HAMMERTIME- A NOTE TO THE G.O.A.T!



I'm writing this from the middle of my last year of college. Hammertime is the thing I keep coming back to between all that — not because anyone asked me to build it, but because it's mine, and because building it is the closest I get to feeling like the work actually means something.

That's why this project exists. I wanted to build something that captures that — the part of F1 that happens in the strategy room, in the what-ifs, in the decisions that look obvious afterward and impossible in the moment it's me trying to get a little closer to the sport I love by actually modeling how it thinks, not just watching the result.

This project won't change anything for you. But it's named after you because some of what's in it — the stubbornness to keep refining something instead of giving up on it, the belief that the work is worth doing even when no one's watching — that's stuff I've taken from watching you race for years.

~Yash Warik.

Why V2 exists

PitWall v1 proved the thesis: an XGBoost-plus-Monte-Carlo engine can turn a grid and a track into a credible probability sheet. It is a prediction tool. It produces a snapshot and stops there.

V2 reframes the project as a simulation, not a forecast — something that behaves the way a strategy room behaves: lap by lap, scenario by scenario, decision by decision. The grid isn't a static input anymore; it's a system that evolves under traffic, weather, incidents and calls made mid-race. The product should feel like the screen behind the wall, not the printout handed to the driver before the lights go out.

OBJECTIVES

- ▶ Move simulation fidelity from a static probability table to a dynamic, lap-resolved race state.
- ▶ Build tooling for comparing outcomes, not just producing one.
- ▶ Give the user a sandbox to test a strategic call before the pit wall has to commit to it.
- ▶ Auto-generate a readable account of what happened in a simulation and why.
- ▶ Build a visual identity that reads as operational tooling — not a student dashboard.

PIT WALL — MODULE PRIORITY

POS	REF	MODULE	STATUS
1	TD-01	Vectorized Race Core	Priority 1
2	TD-06	Command Center UI	Priority 1
3	TD-02	Weekend Intelligence	New System
4	TD-04	Strategy Console	New System
5	TD-03	What-If Studio	New System
6	TD-05	Strategic Debrief	New System

TD-01 and TD-06 run in parallel as the foundation pass.

TD-01 VECTORIZED RACE CORE

CORE REWRITE — PRIORITY 1

Current state — v1's Monte Carlo loop simulates one race at a time in plain Python and draws each driver's finishing position from a static probability table. There is no track, no traffic, no weather, no incidents — just outcomes.

Directive — Rebuild the core as a single vectorized NumPy operation across a (simulations × drivers × laps) tensor. Every lap, the engine updates gaps, applies traffic and dirty-air penalties, rolls overtake probability, adjusts for current driver form, advances the weather state, checks for Safety Car or VSC triggers, and samples pit-stop variance.

SPEC

Tensor shape	(n_simulations, n_drivers, n_laps) — fully vectorized, no per-race Python loop
--------------	--

Traffic / dirty air	Pace penalty applied within a configurable gap window (e.g. <1.0s), scaled by car/track sensitivity
Overtake model	Logistic function of pace delta, DRS availability flag, and tyre-age delta
Driver form	Multiplier derived from rolling championship position and recent-result trend
Weather	3-state Markov chain (dry / intermediate / wet) with circuit-specific transition matrices
Safety Car / VSC	Poisson-distributed incident events drawn from circuit-level historical base rates
Pit stop variance	Time loss sampled from $N(\mu, \sigma)$ calibrated per team
Throughput target	10,000 full-race simulations in under 5 seconds on Render's free tier

TD-02 WEEKEND INTELLIGENCE

NEW SYSTEM

Current state — v1 has no concept of the calendar. Every prediction is generated cold, with no awareness of where the season actually is.

Directive — Build a home view that knows what weekend it is. Auto-detect the active or upcoming race weekend, run live countdowns to each session, and during a session window surface a Live Timing Wall — a simulated telemetry feed seeded from the engine's projected race state, updating on an interval to feel live.

SPEC

Calendar source	Published F1 season schedule (ICS / JSON), synced and cached
Countdowns	Per-session, timezone-aware — FP1–FP3, Sprint, Qualifying, Race
Session state machine	PRE-WEEKEND → LIVE → POST-SESSION
Live Timing Wall	Polling feed (1–5s interval): position, gap to leader/car ahead, last lap, tyre compound + age
Data source	Simulation-projected state — not licensed live telemetry (see Article 7, Flags)

TD-03 WHAT-IF STUDIO

NEW SYSTEM

Current state — v1 returns one outcome distribution per race. There's no way to ask what changes if one variable is different.

Directive — Build a branching scenario engine. The user takes a baseline race, stacks one or more perturbations onto it, and the engine re-simulates from that point forward — sharing the RNG seed with the baseline up to the point of divergence, so the comparison is controlled rather than noise.

SPEC

Perturbation types	SC_REMOVE · GRID_OVERRIDE(driver, position) · WEATHER_AT_LAP(lap, state) · STRATEGY_OVERRIDE(driver, stops)
--------------------	---

Seeding	Shared RNG seed before the divergence lap; independent sampling after
Output	Side-by-side standings: baseline vs. scenario, deltas highlighted per driver
UI affordance	Stackable “scenario chips” — toggle on/off, combine multiple perturbations

EXAMPLE SCENARIOS

“No Safety Car on Lap 32” · “Verstappen starts P10” · “Rain from Lap 25” · “Ferrari pits both cars on Lap 18”

TD-04 STRATEGY CONSOLE

NEW SYSTEM

Current state — There is no path for a user's own strategic call to enter the simulation at all.

Directive — Build a console where the user defines a strategy for one driver — grid slot, tyre compound sequence, number of stops — and runs it against a field on default/historical strategies. Return win probability, podium probability, and a finishing-position distribution for that driver under the chosen plan.

SPEC

Inputs	Driver, starting grid position, compound sequence, stop count
Batch size	1,000–10,000 simulations per console run
Outputs	Win %, podium %, P(top 10) %, finishing-position histogram
Comparison mode	Run 2–3 strategies side by side for the same driver

COMPOUND REFERENCE (FIA CONVENTION)

SOFT	MEDIUM	HARD	INTER	WET
------	--------	------	-------	-----

TD-05 STRATEGIC DEBRIEF

NEW SYSTEM

Current state — Simulation output is numbers only. Nothing interprets the result.

Directive — Build a rule-based report generator that runs after every simulation or scenario and produces a structured debrief: undercut/overcut attempts and their success rate, overtakes executed vs. blocked, position changes attributable to Safety Car or VSC periods, and a composite team performance score. Output as a short narrative summary alongside a metrics table.

SPEC

Undercut/overcut detection	Compare pit-lap timing and resulting position against the rival in the relevant gap window
Overtake log	Per-lap position-change events, tagged by cause: pace / DRS / error / SC-restart
Performance score	Weighted 0–100 composite of pace delta, strategy delta, and race-craft delta vs. season baseline

Output format	3–5 sentence templated narrative + metrics table (rule-based, no LLM dependency required)
---------------	---

TD-06 COMMAND CENTER UI

VISUAL OVERHAUL — PRIORITY 1

Current state — v1's interface is functional but generic — it reads as a demo, not as a tool a team would actually run on a strategy-room screen.

Directive — Redesign the full interface around the visual language of a constructor's strategy room: dense data readouts, monospace telemetry, team-colour driver tags, FIA tyre-compound colour coding, and a low-glare palette built for long sessions. Core components: timing tower, console panels, scenario comparison view, telemetry charts.

SPEC

Component library	Timing tower row, driver tag (team colour + number), tyre chip, gap/delta cell, session countdown, scenario chip
Palette	Neutral base, single high-visibility accent (Ferrari red) for primary actions and alerts; compound colours reserved for data only
Typography	Monospace for all numeric/telemetry data; condensed sans for headings
Responsive behaviour	Timing tower collapses to horizontal scroll below 980px

ARTICLE 6 / SEASON CALENDAR

Sequencing

Ordered as a season, not a backlog — each phase ships a usable increment before the next one starts.

PHASE	REF	DELIVERABLE
Pre-Season Testing	TD-01	Core Engine Rewrite Vectorize the simulation core and calibrate outputs against historical race results before anything is built on top of it.
Round 1–3	TD-06 / TD-02	UI Overhaul & Weekend Intelligence Stand up the command-center shell and calendar awareness in parallel with the engine work, so the new core has somewhere to land.
Round 4–7	TD-04	Strategy Console First user-facing simulation control surface — single-driver strategy testing against the field.
Round 8–10	TD-03	What-If Studio Branching scenarios built on top of the console's batch-run infrastructure.
Round 11+	TD-05	Strategic Debrief & Calibration Pass Narrative layer on top of every simulation, plus a full pass re-validating the engine against the season's actual results.

ARTICLE 7 / FLAGS

Risks & Conditions

YELLOW FLAG	Official live telemetry is licensed and not realistically accessible. The Live Timing Wall (TD-02) must run on simulation-projected state, not real feeds — and should say so.
YELLOW FLAG	10,000-race batches at lap resolution are heavy. The 5-second throughput target on free-tier hosting (TD-01) needs to hold up under TD-03 and TD-04, which both trigger fresh batch runs.
RED FLAG	If simulated probabilities drift from real race outcomes, every downstream module — strategy console, what-if studio, debrief — inherits that error. Calibration against historical results isn't optional polish; it's the gate before TD-02–06 are considered trustworthy.
CHECKERED FLAG	Done means: the engine simulates 10,000 races within target time and within an acceptable error band of historical baselines, all six modules are integrated into one command-center view, and the UI is distinct enough that a strategist would screenshot it without embarrassment.

ARTICLE 8 / SCOPE

In & Out

IN SCOPE — V2	OUT OF SCOPE — THIS REVISION
▸ Vectorized NumPy simulation core ▸ Weekend detection & session countdowns ▸ Simulated live timing wall ▸ Branching what-if scenarios ▸ Single-driver strategy console ▸ Rule-based strategic debriefs ▸ Full command-center UI redesign	– Licensed live telemetry / official data feeds – Multi-user accounts & saved scenario libraries – Native mobile application – Betting / odds integrations – LLM-based debrief generation

HAMMERTIME PROGRAMME — STRATEGY DIVISION
Issued for internal validation only — not for external circulation